NN scattering + phase shifts

NN scattering data, i.e., total + differential cross sections polarizations, analyzing powers... are commonly collected and analyzed in Partial Wave Phase Shifts. These analyses are currently carried out by two major groups: VPI (Arndt + Roper, et al.) and Nijmegen (de Swart et al.). Another group led by Bugg was active in early 1980's, but is now disbanded? In practical terms, PWPSA parameterize a few thousand data points with ~ 150 numbers in some smooth representation. NN potentials are often then fit to these phase shifts — the potentials representing a further reduction to ~ 15-30 parameters. In the end, it is straightforward to test a potential directly against data, i.e. $V_{NN} \rightarrow [\delta_{NN}(^{2S+1}L_J)] \rightarrow \sigma, \frac{d\sigma}{d\Omega} \ldots$ and a χ^2/data number arrived at. χ^2 on the order of $\lesssim 1.5$ are readily achievable with simple parameterization like Argonne

Allowed partial waves $^{2S+1}L_J$

space ⊗ spin ⊗ isospin = antisymmetric

L = even	S = 1	T = 0 (np)	$^3S_1 - {}^3D_1$, $^3D_2$, $^3D_3 - {}^3G_3$, $^3G_4 \ldots$
	S = 0	T = 1 (pp, np, nn)	$^1S_0$, $^1D_2$, $^1G_4 \ldots$
L = odd	S = 1	T = 1	$^3P_0$, $^3P_1$, $^3P_2 - {}^3F_2$, $^3F_3$, $^3F_4 - {}^3H_4$
	S = 0	T = 0	$^1P_1$, $^1F_3$, ...

Schrodinger eqn. for NR scattering:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi + V\psi = E\psi$$

In asymptotic region, $V \to 0$, + for elastic scattering

$$\psi \sim e^{ikz} + f(\theta)\frac{e^{ikr}}{r}$$

unscattered wave scattered wave

$f(\theta)$ = scattering amplitude

In partial wave expansion

$$\psi(r,\theta) = \sum_{l=0}^{\infty} a_l Y_{l0}(\theta) R_l(r) = \sum_{l=0}^{\infty} a_l Y_{l0}(\theta)\frac{u_l(r)}{r}$$

and S.E. reduces to (central potential, no Coulomb)

$$-\frac{\hbar^2}{2\mu}\left\{\frac{d^2u_l}{dr^2} - \frac{l(l+1)}{r^2}u_l\right\} + Vu_l = Eu_l$$

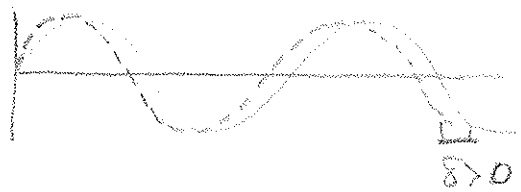
or $$u_l'' = \left\{\frac{l(l+1)}{r^2} + \frac{2\mu}{\hbar^2}V - k^2\right\}u_l \qquad \left(\frac{\hbar^2k^2}{2\mu} = E\right)$$

Now if $V(r) \equiv 0$, i.e., free scattering

$$\frac{u_l(k,r)}{r} \Rightarrow j_l(kr) \to \frac{1}{kr}\sin\left(kr - \tfrac{1}{2}l\pi\right)$$

For $V(r) \neq 0$, but elastic scattering, the probability current density in each partial wave is conserved, so at most, the asymptotic wave function can shift by an amount δ_l

$$\frac{u_l(k,r)}{r} \to \frac{1}{kr}\sin\left(kr - \tfrac{1}{2}l\pi + \delta_l\right)$$



For $\delta > 0$, node is reached for smaller value of r and wave is "sucked in" or a net attractive scattering. If $\delta < 0$, wave is "pushed out" implying net repulsion.

scattering amplitude can be expressed as

$$f(\theta) = \frac{\sqrt{4\pi}}{k} \sum_{l=0}^{\infty} \sqrt{2l+1}\, e^{i\delta_l} \sin\delta_l\, Y_{l0}(\theta)$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \frac{4\pi}{k^2} \left| \sum_l \sqrt{2l+1}\, e^{i\delta_l} \sin\delta_l\, Y_{l0}(\theta) \right|^2$$

and $\sigma = \int \frac{d\sigma}{d\Omega}\, d\Omega = \frac{4\pi}{k^2} \sum_l (2l+1) \sin^2\delta_l$ (at each k)

The cross section at low energy is expressed in terms of a scattering length a and effective range r_e

$$\lim_{k\to 0} \sigma = 4\pi a^2 = 4\pi \frac{\sin^2\delta_l}{k^2}$$

Up to order k^2 the expansion can be made (see Wong sec. C-3)

$$k \cot\delta_0(k) = -\frac{1}{a} + \frac{1}{2} r_e k^2$$

<u>Numerov method</u> (also Cowell or Fox-Goodwin method)
(see D.R. Hartree <u>Numerical Analysis</u> (1958))
To actually compute the phase shift for a given potential, we simply need to solve the S.E. out to a sufficiently large distance and then find the δ that matches the asymptotic ψ

The single-channel S.E. can be written as

$$u_\ell''(r) = g(r)\,u(r) \qquad g(r) = \left\{ \frac{\ell(\ell+1)}{r^2} + \frac{2\mu}{\hbar^2} V(r) - k^2 \right\}$$

A very efficient method for solving $u'' = gu$ is the Numerov method:

Consider a uniform grid in x, spacing δx:
Expand u about point x_0

$$u_+ = u(x+\delta x) = u(x) + \delta x\, u'(x) + \tfrac{1}{2}(\delta x)^2 u''(x) + \tfrac{1}{6}(\delta x)^3 u'''(x) + \tfrac{1}{24}(\delta x)^4 u''''(x)$$
$$u_- = u(x-\delta x) = \quad " \quad - \quad " \quad + \quad " \quad - \quad " \quad + \quad "$$

then the second difference

$$u_+ - 2u_0 + u_- = (\delta x)^2 u_0'' + \tfrac{1}{12}(\delta x)^4 u_0'''' \qquad (1)$$

do the same expansion for $u''(x)$:

$$u_+'' = u_0'' + \delta x\, u_0''' + \tfrac{1}{2}(\delta x)^2 u_0''''$$
$$u_-'' = u_0'' - \delta x\, u_0''' + \tfrac{1}{2}(\delta x)^2 u_0''''$$

$$u_+'' - 2u_0'' + u_-'' = (\delta x)^2 u_0'''' \qquad (2)$$

and substitute l.h.s. of (2) into r.h.s. of (1)

$$u_+ - 2u_0 + u_- = (\delta x)^2\left[u_0'' + \frac{1}{12}\left(u_+'' - 2u_0'' + u_-''\right)\right]$$

now substitute $gu = u''$ on r.h.s.

$$u_+ - 2u_0 + u_- = (\delta x)^2\left[\frac{1}{12}g_+u_+ + \frac{5}{6}g_0u_0 + \frac{1}{12}g_-u_-\right]$$

or $$\left(1 - \frac{1}{12}(\delta x)^2 g_+\right)u_+ = \left(2 + \frac{5}{6}(\delta x)^2 g_0\right)u_0 - \left(1 - \frac{1}{12}(\delta x)^2 g_-\right)u_-$$

so $u_+ \equiv u(x+\delta x)$ is computed knowing u_0, u_- and g_+, g_0, g_-. This is very efficient + precise way of solving S.E. error of order $(\delta x)^4$

To compute the asymptotic wave function, start with $x_- = 0$, $x_0 = \delta x$, $x_+ = 2\delta x$

$u_- = 0$, $u_0 = c$, u_+ to be computed

$(gu)_- = 0$ $(gu)_0 =$ $(gu)_+ =$

even if g infinite!

note $u(r \to 0) \propto r^{l+1}$ (at least).

Propagate u out to a large distance $R \gtrsim 10$ fm beyond range of V. Then compute $u(R)$, $u'(R) = \frac{u(R+\delta x) - u(R)}{2\delta x}$

(see also Chap. 13 of <u>Theoretical Physics on the PC</u>, E.W. Schmid et al)

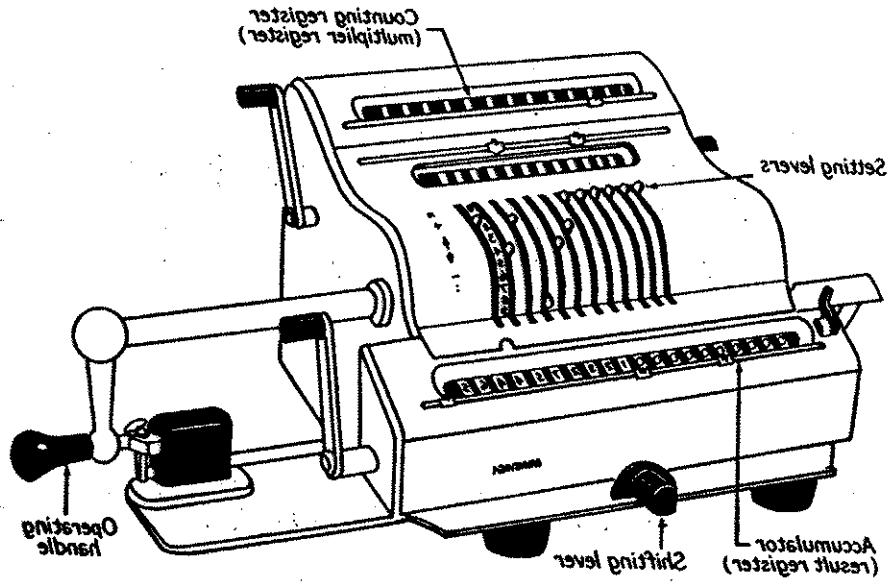


FIG. 1. Brunsviga (hand-operated).

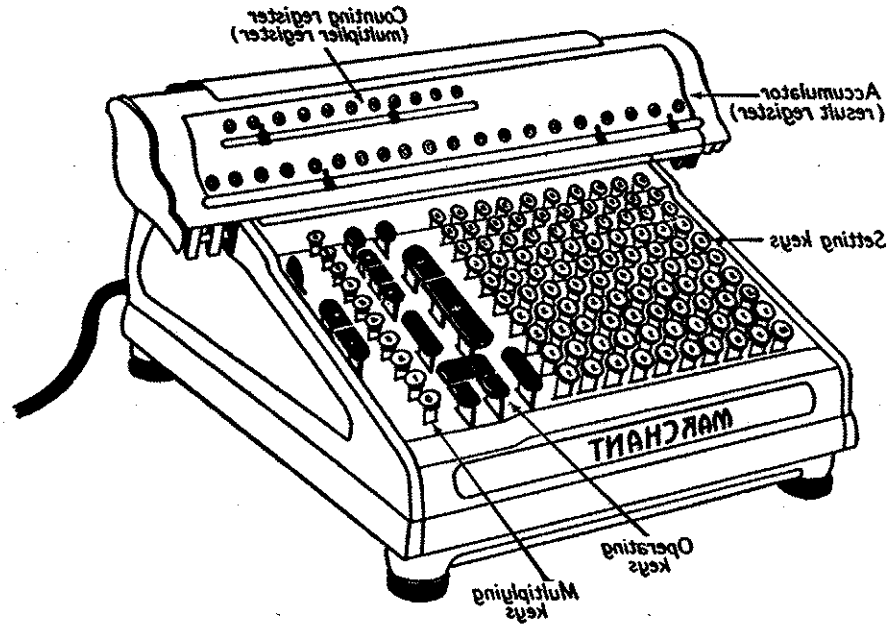


FIG. 2. Marchant (electrically operated).

Now $\frac{u_\ell(k,r)}{r} \propto \frac{1}{kr} \sin(kr - \frac{1}{2}\ell\pi + \delta_\ell)$

$$\approx \cos\delta_\ell \frac{\sin kr}{kr} + \sin\delta_\ell \frac{\cos kr}{kr}$$

$$\frac{u_\ell(kr)}{r} = C\left(\cos\delta_\ell\, j_\ell(\rho) + \sin\delta_\ell\, n_\ell(\rho)\right) \qquad \rho = kr$$

Similarly $\frac{\partial u_\ell}{\partial r} \rightarrow$ function of $\delta_\ell, j_\ell', n_\ell'$

solve for 2 equations in 2 unknowns, C, δ_ℓ by matching to computed $u_\ell(R), u_\ell'(R)$.

Coupled-channels

When there is a tensor force, L is no longer a conserved quantum #, but $J = L+S$ is. The tensor force (present only in $S=1$ states) can connect $L=J-1$ to $L=J+1$ states.

$$\underset{L=0}{\uparrow\uparrow} \longleftrightarrow \underset{L=2}{\downarrow\downarrow}$$

In this case it is common to parameterize the wave function in terms of the S-matrix and 2 phase shifts + 1 mixing parameter

$$\frac{u}{r} \xrightarrow{r\to\infty} \underset{\text{incoming}}{AI} + \underset{\text{outgoing wave}}{A^*O} = A(I + SO) \qquad S = A^{-1}A^*$$

for elastic scattering

with $I = j_\ell(\rho) - i\, n_\ell(\rho)$ $\quad O = j_\ell(\rho) + i\, n_\ell(\rho)$

$$S = \begin{pmatrix} e^{2i\delta_{J>}} \cos 2\varepsilon_J & e^{i(\delta_{J>}+\delta_{J<})} \sin 2\varepsilon_J \\ -e^{i(\delta_{J>}+\delta_{J<})} \sin 2\varepsilon_J & e^{2i\delta_{J<}} \cos 2\varepsilon_J \end{pmatrix}$$

For example, in $^3S_1 - {}^3D_1$ channel, there are $\delta_{J>} = \delta(^3S_1)$, $\delta_{J<} = \delta(^3D_1)$ and ε_1

To obtain scattering lengths, solve $^1S_0$ or $^3S_1 - {}^3D_1$ for a collection of low-energies $\lesssim 1\,\mathrm{MeV}$ and fit to form

$$k \cot\delta = -\frac{1}{a} + \frac{1}{2} r_e k^2$$

Coulomb complications

In pp scattering we need to include the long-range Coulomb potential in V, and match onto Coulomb wave functions, rather than Bessel functions asymptotically:

$$\frac{d^2 u}{d\rho^2} = \left\{ \frac{\ell(\ell+1)}{\rho^2} + \frac{2\eta}{\rho} - 1 \right\} u \qquad \rho = kr \quad k = \frac{mv}{\hbar} \quad \eta = \frac{\alpha}{v}$$

$$\alpha = \frac{e^2}{\hbar c}$$

The regular and irregular solutions are the Coulomb functions F_ℓ, G_ℓ. They play a role corresponding to: $\rho j_\ell(\rho) \leftrightarrow F_\ell$ $\quad -\rho n_\ell(\rho) \leftrightarrow G_\ell$

Standard subroutines available for $F_\ell, G_\ell(\rho, \eta)$

Other than inserting potential in S.E. and matching $\mathcal{U}_\ell(R)\mathcal{U}_\ell'(R)$ to $F_\ell(R)$, ..., exact same procedure for computing δ as for np (or nn) scattering.

Scattering length is more complicated:

$$C^2 k \cot\delta + \frac{1}{R} h(\eta) = -\frac{1}{a} + \frac{1}{2} r_e k^2 - P r_e^3 k^4 + \ldots$$

where $R = \frac{\hbar^2}{m_p e^2}$ (Coulomb length) $\quad C^2 = \frac{2\pi\eta}{e^{2\pi\eta}-1}$

$$h(\eta) = \eta^2 \sum_{m=1}^{\infty} \frac{1}{m(m^2+\eta^2)} - \ln\eta - \gamma \qquad \gamma = 0.57722\ldots \text{ (Euler's const.)}$$

Again, fit collection of computed δ's (down to $E \sim .05$ MeV) to form to extract a, r_e

Comparing $^1S_0(pp)$ to $^1S_0(np)$ we see differences in $\delta(1\text{MeV}) \sim 20^\circ$ $\quad \delta(5\text{MeV}) \sim 5^\circ$ from $\frac{e^2}{r}$

Other corrections for precise comparison to data $\lesssim 1$ MeV include vacuum polarization

and magnetic moment interaction

these are corrections $\sim 0.1^\circ$ at 1 MeV

Deuteron

The measured properties of the deuteron include E_d, Q_d, μ_d, r_d, A_s, $\eta = A_D/A_s$, $(J^\pi, T) = (1^+, 0)$. Symmetry arguments lead to the conclusion that the wave function is a $^3S_1 - {}^3D_1$ admixture, i.e. $S=1$, $T=0$, with $L=0$ or 2, total $\vec{J} = \vec{L} + \vec{S} = 1$. The D-wave admixture is necessary to get $Q_d \neq 0$, and helps explain why $\mu_d \neq \mu_p + \mu_n$. It is also expected from OPE interactions and tensor force they produce.

Having convinced ourselves of these general arguments, how do we actually solve for the deuteron wave function, given a realistic V_{NN}?

For $V_{NN} = V_c + V_\sigma \sigma_1 \cdot \sigma_2 + V_T S_{12} + V_{LS} \vec{L} \cdot \vec{S}$

with $V_i = V_i^0(r) + V_i^\tau(r)\, \tau_1 \cdot \tau_2$ we note that $\tau_1 \cdot \tau_2 |T=0\rangle = -3|T=0\rangle$, $\sigma_1 \cdot \sigma_2 |S=1\rangle = 1|S=1\rangle$ so in the deuteron,

$$V_{NN} = \tilde{V}_c(r) + \tilde{V}_T(r) S_{12} + \tilde{V}_{LS}(r)\, \vec{L} \cdot \vec{S}$$

with
$$\tilde{V}_c(r) = V_c^0(r) - 3V_c^\tau(r) + V_\sigma^0(r) - 3V_\sigma^\tau(r)$$
$$\tilde{V}_T(r) = V_T^0(r) - 3V_T^\tau(r)$$
$$\tilde{V}_{LS}(r) = V_{LS}^0(r) - 3V_{LS}^\tau(r)$$

The deuteron wave function may be written

$$\psi_d = \frac{1}{r}\left[u(r)\,\mathcal{Y}^1_{1,10} + w(r)\,\mathcal{Y}^1_{1,12}\right]$$

$$\mathcal{Y}^M_{J,SL} = \sum_{m_L+m_S=M} C(L\,S\,m_L\,m_S\,|\,J\,M)\,Y_{Lm_L}(\theta,\phi)\,\chi_{Sm_S}$$

so $\mathcal{Y}^1_{1,10} = Y_{00}(\theta,\phi)\,\chi_{11}\ \left(= \frac{1}{\sqrt{4\pi}}\,|\uparrow\uparrow\rangle\right)$

$$\mathcal{Y}^1_{1,12} = \sqrt{\tfrac{3}{5}}\,Y_{22}\,\chi_{1-1} - \sqrt{\tfrac{3}{10}}\,Y_{21}\,\chi_{10} + \sqrt{\tfrac{1}{10}}\,Y_{20}\,\chi_{11}$$

Need to know: $\langle \mathcal{Y}^M_{J,1L}\,|\,S_{12}\,|\,\mathcal{Y}^M_{J,1L'}\rangle$

	$L' = J-1$	J	$J+1$
$L = J-1$	$-2(J-1)/(2J+1)$	0	$6\sqrt{J(J+1)}/(2J+1)$
J	0	2	0
$J+1$	$6\sqrt{J(J+1)}/(2J+1)$	0	$-2(J+2)/(2J+1)$

also $\langle \mathcal{Y}^M_{J,1L}\,|\,\vec{L}\cdot\vec{S}\,|\,\mathcal{Y}^M_{J,1L'}\rangle$

	$L' = J-1$	J	$J+1$
$L = J-1$	$J-1$	0	0
J	0	-1	0
$J+1$	0	0	$-(J+2)$

so $S_{12}\,\mathcal{Y}^1_{1,10} = \sqrt{8}\,\mathcal{Y}^1_{1,12}$ $\qquad S_{12}\,\mathcal{Y}^1_{1,12} = \sqrt{8}\,\mathcal{Y}^1_{1,10} - 2\,\mathcal{Y}^1_{1,12}$

$L\cdot S\,\mathcal{Y}^1_{1,10} = 0$ $\qquad L\cdot S\,\mathcal{Y}^1_{1,12} = -3\,\mathcal{Y}^1_{1,12}$

order to account for the observed spin of 1, and the magnetic moment $0.82\mu_N$. The nucleon-nucleon interaction is assumed to be of the form $mP+nQ$, where P denotes the Majorana and Q the Bartlett interaction. In order to get some indication of the effect of the form of the nuclear interaction on the result, calculations have been carried out for various types of potential and for various "ranges," exponential, Yukawa, and Gaussian. The constants m, n, and depth of the potential have been estimated from data for two free nucleons. The secular equations are solved by perturbation methods, the only unknown parameter being the spin-orbit interaction parameter ζ. The energy levels for the various types of interaction and "ranges" have been plotted against ζ, whose value is then determined from known ground and excited state of Li^6. The coupling is found to be much closer to $L\cdot S$ than to $j\cdot j$. The magnetic moment so calculated is about 0.87, in much better agreement with the experimental value than that obtained from extreme $j\cdot j$ coupling.

* Postdoctorate Fellow.

EA10. Magnetic Moment Deviations from the Schmidt Limits via Admixed States. A. B. VOLKOV, *University of Wisconsin.*—It has been noticed[1] that the relative deviations of the experimental magnetic moments from the Schmidt limits are nearly the same for pairs of odd-even nuclei of equal spin J, if the one has the same number (n) of odd protons as the other has neutrons. This result can be understood on the basis of two simple assumptions: that the angular momentum is distributed only among the n (odd) particles and that these particles satisfy a mirror condition between the two nuclei. Then the calculated relative deviations from the Schmidt lines are identical for any admixture of odd particle states. The experimental evidence may therefore be taken as an indication that, to a good approximation, the angular momentum is shared only among the n (odd) particles. If just this assumption is applied to any odd-even nucleus, one may obtain a simple expression for its magnetic moment as a function of the squared amplitudes of the Russell-Saunders terms of the admixed states. These results may be used in conjunction with the observed deviations to study the nature of the admixed states.

[1] A. L. Schawlow and C. H. Townes, Phys. Rev. 82, 268 (1951).

EA11. Contributions to the Hyperfine Spectrum of He^3 Resulting from the Nuclear Structure. A. M. SESSLER* AND H. M. FOLEY, *Columbia University.*—The recent experimental determination[1] of the hfs splitting in the ^2S state of He^3 makes it important to compute the specific effects of the nuclear structure and the nucleon exchange currents. In first approximation it may be shown that the electrons near the nucleus move in orbits determined by the instantaneous position of the two protons, as in the case of deuterium.[2] In this region a one electron treatment may be employed, and advantage is taken of the separability of the two center (H_2 ion) in the relativistic problem. Relativistic corrections are then made. Calculation of the absolute value of the hfs splitting requires an accurate value of the electron charge density at the nucleus. Dr. P. J. Luke is engaged in this calculation.[3] The present work is expressed as a relative correction to the hfs as calculated for a point nucleus. The nuclear wave function of Pease and Feshbach[4] is used to determine the spatial distribution of the nucleons and the orbital hfs. The contribution of exchange currents to the hfs according to various models is being investigated.

* National Science Foundation Predoctoral Fellow.
[1] G. Weinreich and V. Hughes, Meeting of American Physical Society, Cambridge, Massachusetts (1953).
[2] A. Bohr, Phys. Rev. 73, 1109 (1948); F. Low, Phys. Rev. 77, 361 (1950).
[3] P. Luke (private communication).
[4] R. Pease and H. Freshbach, Phys. Rev. 88, 945 (1952).

EA12. The Ground State of the Deuteron with Tensor Forces. I. M. H. KALOS AND J. M. BLATT, *University of Illinois.*—A program has been prepared for the Illinois high speed digital computer to solve the following problem: The potential acting in the ground state of the deuteron is written in the form $V=V_{c1}(r)+\lambda[V_{c2}(r)+V_t(r)S_{12}]$, where S_{12} is the conventional tensor operator and the three functions V are arbitrary (i.e., their numerical values at the netpoints used for integration of the differential equations are at the disposal of the programmer). An iteration method[1] is used which converges onto that value of λ which leads to the correct binding energy of the deuteron. The wave functions $u(r)$ (S state) and $w(r)$ (D state) are found during this iteration. The machine then computes and prints out the quadrupole moment, D state probability, and triplet state effective range implied by the potential, as well as a complete table of the wave functions u and w at each netpoint, normalized to $\int_0^\infty(u^2+w^2)dr=1$. The computation takes between 5 and 20 minutes, the printing takes another 5 minutes.

[1] Herman Feshbach and Julian Schwinger, Phys. Rev. 84, 194 (1951).

EA13. The Ground State of the Deuteron with Tensor Forces. II. J. M. BLATT AND M. H. KALOS, *University of Illinois.*—The computing program described in the previous abstract has been used to investigate the dependence of the ground state parameters of the deuteron (quadrupole moment, D state probability) upon various features of the potential (range, depth, ratio of tensor depth to central force depth, ratio of the ranges, shapes). This has been done for conventional potential wells as well as for repulsive-core wells. The results will be discussed.

THURSDAY AFTERNOON AT 2:00

Shoreham, Terrace Room

(R. E. MARSHAK presiding)

Cosmic Rays: Stars, Curious Particles

F1. Cosmic-Ray Star Production Rates in Gases.* FIELDING BROWN† AND GEORGE T. REYNOLDS, *Princeton University.*—Extending earlier cloud-ion chamber studies of stars in neon, argon, and krypton,[1] the rates of star production at mountain altitude have been determined for stars of $\geqslant 1$ prong and for $\geqslant 3$ prongs. In the former case the star rate in Ne^{20} relative to that in Kr^{83} exceeds that expected from relative nuclear geometric areas by a factor of between 5 and 7. In the latter case the relative rate is 2.9 times the geometrical expectation. Rates in A^{40} are intermediate. Since large corrections have been necessary for those stars of 1 and 2 prongs missed due to instrumental bias; but little correction for stars $\geqslant 3$ prongs, the rates for $\geqslant 3$ prongs are considered more reliable. Such an effect is supported by earlier emulsion data at this altitude.

Phys. Rev. 91 (1953)

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi_d + V\psi_d = E\psi_d \equiv -B\psi_d \qquad B>0$$

after some manipulation becomes pair of coupled equations (essentially 1 for ψ'_{110} + 1 for ψ'_{112})*

$$-\frac{\hbar^2}{2\mu}u'' + (\tilde{V}_c + B)u + \sqrt{8}\tilde{V}_T w = 0$$

$$-\frac{\hbar^2}{2\mu}\left(w'' - \frac{6}{r^2}w\right) + \left(\tilde{V}_c - 2\tilde{V}_T - 3\tilde{V}_{LS} + B\right)w + \sqrt{8}\tilde{V}_T u = 0$$

or $\quad U'' = GU \quad$ for $\quad U = \begin{pmatrix} u \\ w \end{pmatrix}$

$$G = \begin{pmatrix} \frac{2\mu}{\hbar^2}(\tilde{V}_c + B) & \sqrt{8}\,\frac{2\mu}{\hbar^2}\tilde{V}_T \\ \sqrt{8}\,\frac{2\mu}{\hbar^2}\tilde{V}_T & \frac{6}{r^2} + \frac{2\mu}{\hbar^2}\left(\tilde{V}_c - 2\tilde{V}_T - 3\tilde{V}_{LS} + B\right) \end{pmatrix}$$

Now at large distance, $V_{NN} \to 0$ and equations are uncoupled, with solutions:

$$U \to \begin{pmatrix} A_S e^{-\gamma r} \\ A_D\left(1 + \frac{3}{\gamma r} + \frac{3}{\gamma^2 r^2}\right)e^{-\gamma r} \end{pmatrix} = \begin{pmatrix} A_S f_S(r) \\ A_D f_D(r) \end{pmatrix}$$

At $r=0$, $U \to \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\qquad$ where $\frac{\hbar^2\gamma^2}{2\mu} = B$

* remember $\nabla^2 = \frac{1}{r}\frac{\partial^2}{\partial r^2}r - \frac{L^2}{r^2}$

To solve, use Numerov in a matrix form, for two independent solutions:

$$\phi(r=0)=\begin{pmatrix}0\\0\end{pmatrix} \quad \phi(r=\delta x)=\begin{pmatrix}c\\0\end{pmatrix} \quad \psi(0)=\begin{pmatrix}0\\0\end{pmatrix} \quad \psi(\delta x)=\begin{pmatrix}0\\c\end{pmatrix}$$

$$\left[1-\frac{\delta x^2}{12}G_+\right]U_+=\left[2+\frac{5}{6}\delta x^2 G_0\right]U_0-\left[1-\frac{\delta x^2}{12}G_-\right]U_-$$

$$\text{or } U_+=\left[1-\frac{\delta x^2}{12}G_+\right]^{-1}\left(\qquad\qquad\qquad\qquad\right)$$

$$\left[1-\frac{\delta x^2}{12}G\right]^{-1}=\begin{pmatrix}1-\frac{\delta x^2}{12}G_{cc} & -\frac{\delta x^2}{12}G_{ct}\\ -\frac{\delta x^2}{12}G_{tc} & 1-\frac{\delta x^2}{12}G_{tt}\end{pmatrix}^{-1}$$

$$=\frac{1}{\det G}\begin{pmatrix}1-\frac{\delta x^2}{12}G_{tt} & \frac{\delta x^2}{12}G_{ct}\\ \frac{\delta x^2}{12}G_{tc} & 1-\frac{\delta x^2}{12}G_{cc}\end{pmatrix}$$

Given a potential with unknown B, guess a value for B and propagate two independent solutions ϕ, χ out to $r=R \approx 10\text{-}20\,\text{fm}$ beyond range of V. At $r=R$ obtain ϕ, χ, ϕ', χ'.

Now $U=a\phi+b\chi$, $U'=a\phi'+b\chi'$ at $r=R$

$$\text{or}\begin{pmatrix}\phi_c & \chi_c & -f_s & 0\\ \phi_t & \chi_t & 0 & -f_D\\ \phi_c' & \chi_c' & -f_s' & 0\\ \phi_t' & \chi_t' & 0 & -f_D'\end{pmatrix}\begin{pmatrix}a\\ b\\ A_s\\ A_D\end{pmatrix}=0=M\,v$$

will have solutions if $\det(M)=0$

so follow an iterative procedure:

1) pick $B \Rightarrow f_S, f_D, f_S', f_D'$
2) generate two independent solutions to $u'' = Gu$, ϕ and χ (eqn depends on B) out to $r = R$
3) evaluate $\det(M)$
4) adjust B and repeat, until $\det(M) = 0$

now you know solutions exist, and can find a, b, A_D in terms of A_S, and make appropriate linear combination to generate u, w, and to normalize:

$$\int_0^\infty u^2 + w^2 \, dr = 1 \iff \langle \psi | \psi \rangle = 1$$

and remember that ψ is very long ranged, so include

$$\int_R^\infty f_S^2 \, dr = \frac{e^{-2\gamma R}}{2\gamma} \qquad \int_R^\infty f_D^2 = \frac{e^{-2\gamma R}}{2\gamma}\left(1 + \frac{6}{\gamma R} + \frac{12}{(\gamma R)^2} + \frac{6}{(\gamma R)^3}\right)$$

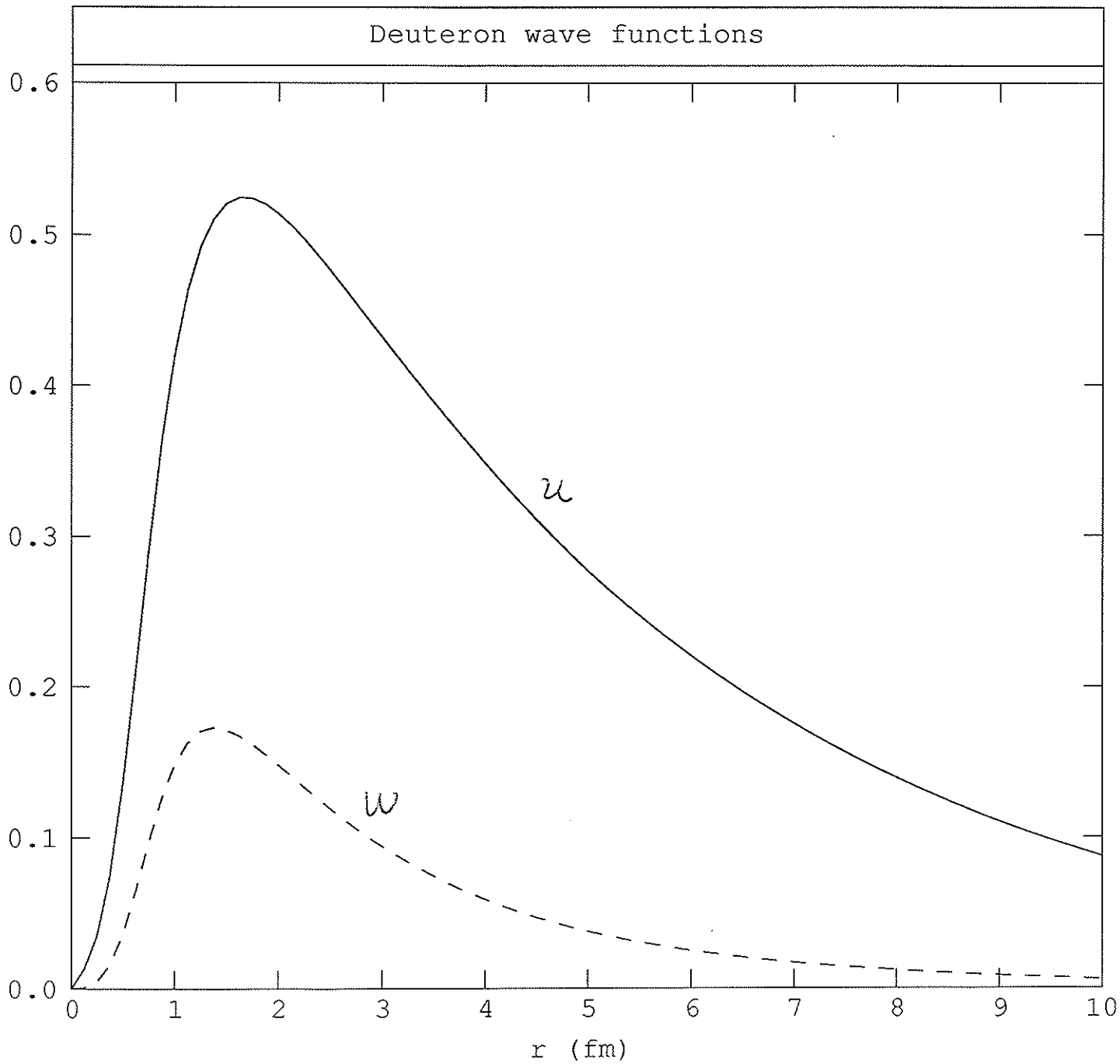
Then you have E_d, A_S, $\eta = A_D/A_S$, $\int_0^\infty w^2 dr = P_D$

Other properties of interest: $r_d = \langle (\frac{r}{2})^2 \rangle^{1/2}$

$$Q_d = \langle e(3z^2 - r^2) \rangle \qquad \mu_d = \langle g_p \vec{S}_p + g_n \vec{S}_n + \tfrac{1}{2}\vec{L} \rangle$$

$$\Rightarrow \mu_d = (\mu_p + \mu_n)\left(1 - \tfrac{3}{2} P_D\right) + \tfrac{3}{4} P_D$$

$$Q_d = \frac{1}{20} \int_0^\infty (\sqrt{8}\, u w - w^2) r^2 dr$$


Deuteron wave functions
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
1
2
3
4
5
6
7
8
9
10
r (fm)
u
w